

SALSABIL OUNI

Sousse, Tunisia . salsabil.ouni.i@gmail.com . +216 54749984 . [linkedin.com/in/salsabil-ouni/](https://www.linkedin.com/in/salsabil-ouni/)

PROFESSIONAL EXPERIENCE

Farhat Hached Hospital <i>Structural Genomics and Protein Bioinformatics Researcher</i>	2026/2 - Present Sousse, Tunisia
- Built a pipeline linking experimental $\Delta\Delta G$ data to FoldX alanine-scan hotspot patches on p53, showing destabilizing variants are enriched in structural microenvironments and replicating this across independent datasets. - Ensured rigor via confounder-adjusted regression, distance/threshold-sensitivity analyses, and cross-assay replication; validated by recovering p53's known functional patches in PyMOL.	
Arcadia <i>Machine Learning Engineer Intern</i>	2025/7 - 2025/9 Osaka, Japan
- Developed an Arabic Text-to-Speech system using VITS (Variational Inference with adversarial learning), combining Conditional VAE and GAN-based architectures to generate natural speech with human-like prosody achieving 97% intelligibility based on Spectrogram analysis - Deployed a real-time speech synthesis platform using Python, Svelte, and Docker with <1.5s latency.	
PURA Solutions <i>Artificial Intelligence Engineer Intern</i>	2024/6 - 2024/10 Sousse, Tunisia
- Built a medical AI system for Thoracic X-ray analysis and report generation using CNN and Transformer architectures, achieving 92% validation performance. - Developed an interactive deployment pipeline using Streamlit for model evaluation and visualization.	
Systems Nakashima <i>Machine Learning and Full Stack Research Engineer Intern</i>	2024/2 - 2024/5 Tokyo, Japan
- Developed Genki Walk, an AI rehabilitation system used by 36+ patients, combining MediaPipe pose estimation and Transformer models for gait analysis. - Built and deployed a healthcare platform using Django, Vue.js, and AWS, achieving 91% gait analysis accuracy.	
Systems Nakashima <i>Artificial Intelligence and Full Stack Engineer Intern</i>	2023/8 - 2023/10 Tokyo, Japan
- Developed Aintone, an AI-powered sales forecasting platform using PyTorch based machine learning models for predictive analytics and built an end-to-end application integrating Django backend and Vue.js frontend. - Presented the project at Tokyo Big Sight 2023, attracting 400+ attendees and receiving positive feedback from industry professionals.	

EDUCATION

Faculty of Medicine Ibn El Jazzar of Sousse <i>Master's Degree in Molecular Medicine : AI Applied to Genetics, Immunology and Microbiology</i>	2024 - Present
Polytechnic School of Sousse <i>Engineering Diploma in Artificial Intelligence and Data Science</i>	2024 - Present
Higher Institute of Applied Science and Technology of Sousse <i>Bachelor of Computer Science and Software Engineering,</i>	2021 - 2024

ADDITIONAL INFORMATION

Technical Skills: Python | Biopython | PyTorch | R | Conda | AlphaFold | FoldX | PyMOL | Modeller | Java | SciPy | Scikit-learn | Transformers | Computer Vision | LLMs| NLP| SQL | PostgreSQL | Linux/WSL | Bash | Docker | Git | AWS | C/C++ | RAG | Kubernetes | GitHub Actions | GitLab CI | Pandas | NumPy | Spark | Tableau | Power BI | Hadoop

Languages: English | French | Arabic | Japanese | German

Certifications: NVIDIA Fundamentals of Deep Learning | NVIDIA Introduction to Transformer-Based Natural Language Processing | NVIDIA Generative AI with Diffusion Models | Microsoft Azure Data Fundamentals

Achievements: AIESEC Waseda Japan Global Talent 2023, selected out of 22 countries globally | AI Project Exhibitor at Tokyo Big Sight 2023 | Winner of 16 Hackathons.

Projects: Protein Secondary Structure Prediction | Gene Expression Exploration | DNA Sequence Classification | Breast Cancer Detection in Women | Protein Thermostability Prediction

Research Interests: AI-driven drug discovery and design | Machine learning applications in pharmaceutical research

Research Papers In Progress: Assessing the Validity of 3D Structural Patches for the Classification of Point Mutations | ML-Based Gait Analysis for TKA Rehabilitation | NLP and ML Approaches to the Endometriosis-Fibromyalgia Link

Volunteering: TEDx Marketing Manager, Hadrumet (2025) | Global Ambassador, IEEE Day Region 8 (2022)